

- + . **Is Poverty a Predictor of Low**
- o **Literacy Rates?**

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Introduction



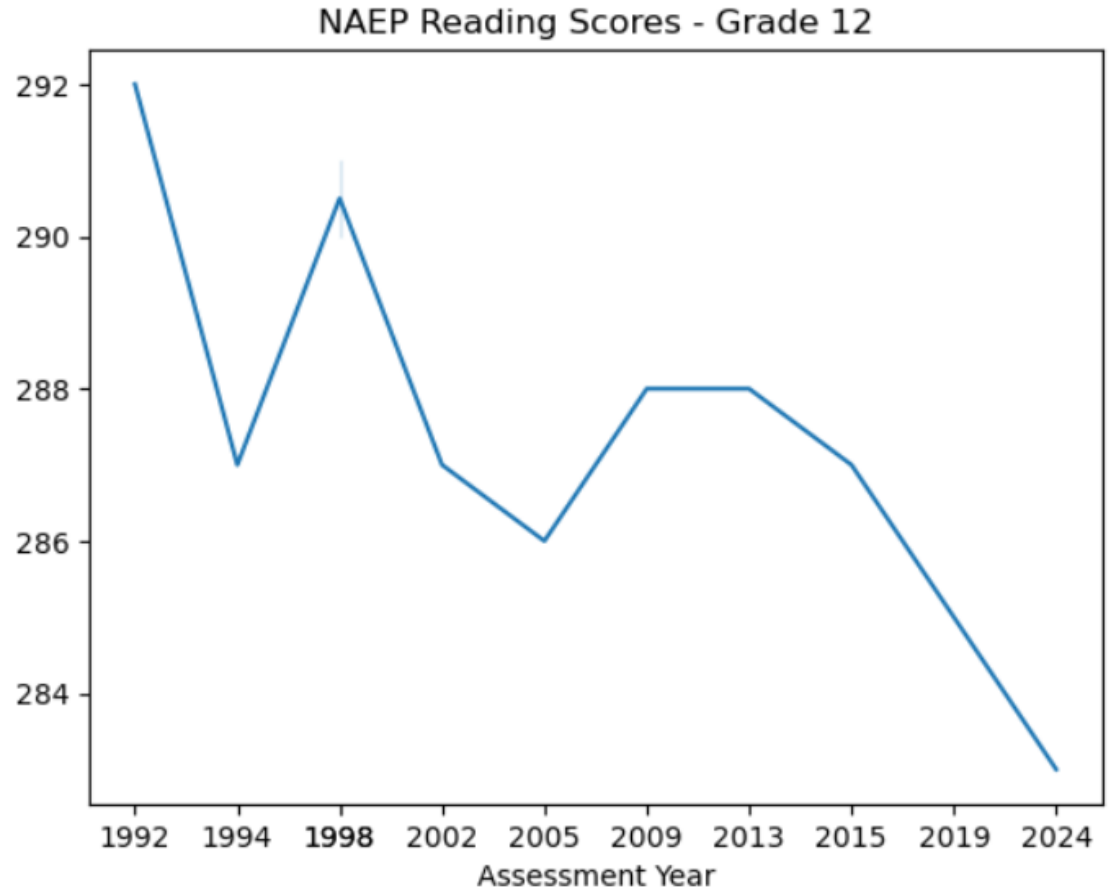
Who am I?

- Sean Bush, soon to be M.S. in Data Science
- Worked as a college math tutor and a high school teacher for several years. Deeply passionate about education
- Developed an interest in data, computing, and creative-problem solving which led to pursuing data science
- Looking to leverage communication and data science skills to aid organizations



The Literacy Decline

- The National Assessment of Education Progress (NAEP) is a periodic assessment of academic skills conducted by the U.S. Department of Education (National Center for Education Statistics, n.d.-b).
- The most recent assessment of senior reading scores in 2024 was the lowest ever recorded by the NAEP (National Assessment Governing Board, 2024).
- This continues a general decline in senior reading scores (National Center for Education Statistics, n.d.-a).



The Question

- Educators and lawmakers need data-backed answers about and solutions to the literacy decline.
- One possible avenue of research would be determining if poverty is a significant contributing factor to low literacy rates.
- Research Question: Is the poverty rate of a county in the U.S., the proportion of residents who live below the poverty line, a statistically significant predictor of that county's literacy score?
- Null Hypothesis: Poverty rate is not a statistically significant predictor of literacy scores.
- Alternative Hypothesis: Poverty rate is a statistically significant predictor of literacy scores.

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• Data Gathering & Preparation



The Data

- The Programme for the International Assessment of Adult Competencies (PIAAC) tests literacy and numeracy across various countries and releases the gathered data as the Skills Map (National Center for Education Statistics, n.d.-c).
- For each county in the U.S., this data set records an average literacy score, the poverty rate, and a variety of other statistics such as the population, employment rate, and demographic makeup (PIAAC, 2023).
- The Skills Map data is publicly available from the Department of Education's website (PIAAC, 2023).

Data Gathering

- The Skills Map website provides the data as a downloadable Excel spreadsheet called “SAE_website_dataset.” It contains multiple worksheets, including one which records the county-level data.
- The data set includes data on each county’s estimated literacy as well as various statistics and demographics.

	A	B	C	D	E	F	
1	FIPS_code	State	County	grpName	Lit_P1	Lit_P1_CI_L	Lit_
2	1001	Alabama	Autauga Cour	all	0.205	0.169	
3	1003	Alabama	Baldwin Coun	all	0.161	0.125	
4	1005	Alabama	Barbour Coun	all	0.394	0.343	
5	1007	Alabama	Bibb County	all	0.269	0.228	
6	1009	Alabama	Blount County	all	0.248	0.204	
7	1011	Alabama	Bullock Count	all	0.447	0.383	
8	1013	Alabama	Butler County	all	0.32	0.274	
9	1015	Alabama	Calhoun Cour	all	0.25	0.215	
10	1017	Alabama	Chambers Cou	all	0.299	0.254	
11	1019	Alabama	Cherokee Cou	all	0.253	0.209	
12	1021	Alabama	Chilton Count	all	0.262	0.219	
13	1023	Alabama	Choctaw Cou	all	0.318	0.273	
14	1025	Alabama	Clarke County	all	0.322	0.269	
15	1027	Alabama	Clay County	all	0.318	0.266	
16	1029	Alabama	Cleburne Cou	all	0.279	0.228	
17	1031	Alabama	Coffee Count	all	0.224	0.189	
18	1033	Alabama	Colbert Count	all	0.233	0.196	
19	1035	Alabama	Conecuh Cou	all	0.355	0.304	
20	1037	Alabama	Coosa County	all	0.32	0.263	
21	1039	Alabama	Covington Co	all	0.243	0.204	
22	1041	Alabama	Crenshaw Cou	all	0.284	0.241	
23	1043	Alabama	Cullman Coun	all	0.226	0.187	
24	1045	Alabama	Dale County	all	0.229	0.19	
25	1047	Alabama	Dallas County	all	0.37	0.313	
26	1049	Alabama	DeKalb Count	all	0.319	0.272	
27	1051	Alabama	Elmore Count	all	0.215	0.177	
28	1053	Alabama	Escambia Cou	all	0.309	0.267	

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County

State

Nation

Variable List

+

Selected Variables

- The full list of variables used in this analysis are as follows:
 - Average literacy score (the outcome variable)
 - Population
 - Proportion of residents who are female
 - Proportion of residents who are black
 - Proportion of residents who are Hispanic
 - Proportion of residents who are Asian
 - Proportion of residents who are American Indian/Alaskan Native
 - Proportion of residents who are Native Hawaiian/Pacific Islander
 - Proportion of residents who are of another non-white race
 - Proportion of residents who are foreign-born
 - Proportion of residents who are below the poverty line
 - Proportion of residents who are employed
 - Proportion of residents who are in the labor force but are currently unemployed
 - Proportion of residents who are in managerial occupations
 - Proportion of residents who are in service occupations
 - Proportion of residents who are in sales occupations
 - Proportion of residents who are in military occupations
 - Proportion of residents who are in production occupations
- Variables were carefully selected so as not to introduce collinearity-correlation between variables.

Data Cleaning

- The data set was exceptionally clean in its published form. There were no missing values in the data set to be handled.
- Outliers were identified using the interquartile range and then examined to determine their validity. No invalid outlying values were found in the data set.
- All predictors were standardized (shifted and scaled such that the standard deviation is 1 and the mean is 0). This ensures all predictors are on the same scale.



The Analysis



Linear Regression at a Glance

- Linear Regression finds the best-fitting line to represent the relationship between two variables
- The Ordinary Least Squares (OLS) method finds the line that minimizes the combination of all differences between the line and actual values
- Multiple linear regression extends this idea to a larger number of variables, where one variable, called the outcome, is listed as a linear combination of the other variables, called predictors.

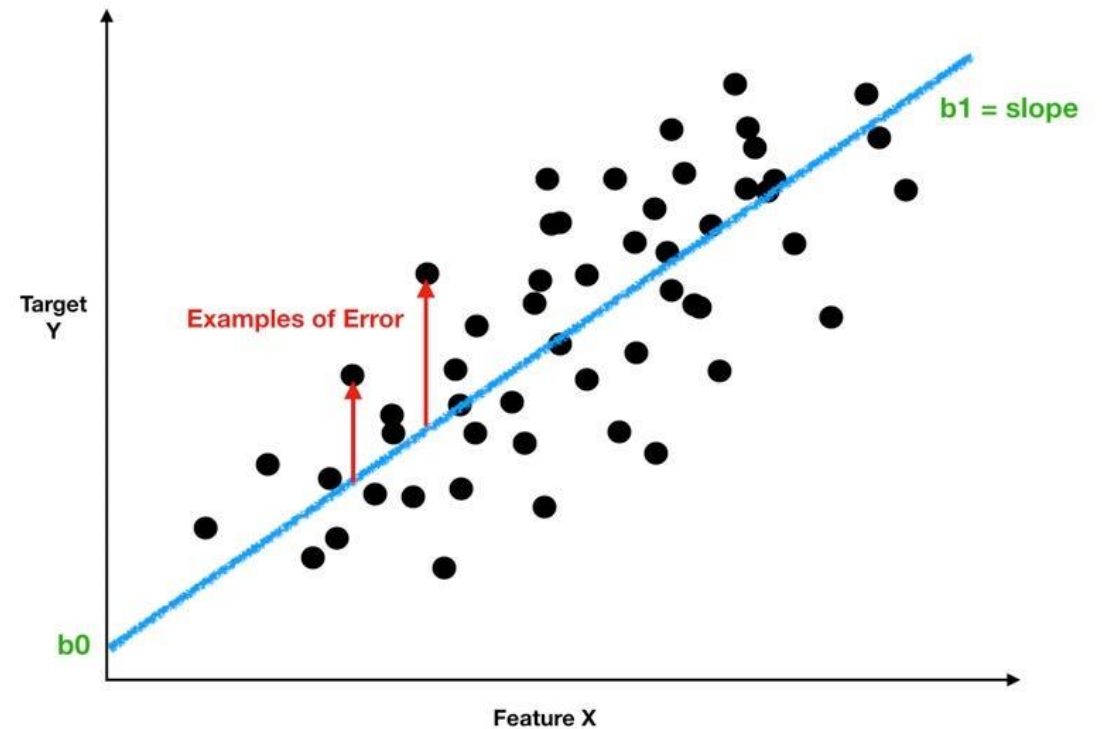


Image source: (Yiu, 2020)

Model Training

- For this analysis, a Linear Regression model was created in Python using the StatsModels package to predict average literacy score as a function of the selected variables listed previously.
- Each variable gets a coefficient that describes how much of an impact it will have on the literacy score. The model StatsModels creates will display these coefficient and p-values associated with the likelihood of the coefficients being nonzero in reality. These will be used to answer the research question.
- The data set was split into training data and testing data, and the model was fit onto the training set.

Model Fit

- R-squared is a measure of how much variance in the outcome variable is explained by the model. In laymen's terms, it is one measure of how well the model describes the data.
- The fitted model achieved an R-squared score of 0.881 on the training data, indicating a well-fit model.
- On the testing set, the model achieved a comparable R-squared score of 0.875, indicating that the model is not overfit on the training data.

OLS Regression Results

Dep. Variable:	avg_literacy	R-squared:	0.881
Model:	OLS	Adj. R-squared:	0.881
Method:	Least Squares	F-statistic:	1091.
Date:	Tue, 21 Apr 2026	Prob (F-statistic):	0.00
Time:	21:29:31	Log-Likelihood:	-7425.5
No. Observations:	2513	AIC:	1.489e+04
Df Residuals:	2495	BIC:	1.499e+04
Df Model:	17		
Covariance Type:	nonrobust		

The Results

- Recall that the goal of the analysis is to establish that poverty rate is a predictor of literacy score. A nonzero coefficient would confirm this hypothesis.
- The model reported a coefficient of -3.212 for the poverty rate variable, indicating a negative correlation.
- The associated p-value of nearly zero, indicates a statistically significant relationship.

	coef	std err	t	P> t	[0.025	0.975]
const	259.9974	0.093	2793.144	0.000	259.815	260.180
population	-0.1221	0.106	-1.151	0.250	-0.330	0.086
female	-0.0745	0.121	-0.614	0.539	-0.312	0.163
black	-3.7722	0.118	-31.963	0.000	-4.004	-3.541
hispanic	-4.9669	0.163	-30.392	0.000	-5.287	-4.646
asian	0.6898	0.171	4.023	0.000	0.354	1.026
native_american	-0.4985	0.113	-4.415	0.000	-0.720	-0.277
native_hawaiian_pacific_islander	-0.2092	0.193	-1.084	0.278	-0.588	0.169
other_race	0.5793	0.122	4.759	0.000	0.341	0.818
foreign_born	-0.6154	0.193	-3.190	0.001	-0.994	-0.237
below_poverty_line	-3.2120	0.170	-18.907	0.000	-3.545	-2.879
employed	3.2053	0.172	18.668	0.000	2.869	3.542

Findings

- The exact p-value associated with the poverty rate variable is 1.29×10^{-74} . This is well below a standard significant level of 0.05 that would be used to rule out the possibility of non-correlation.
- Therefore, the null hypothesis can be rejected in favor of the alternative. The poverty rate of a county in the U.S. is indeed a statistically significant predictor of that county's average literacy score.



Conclusions



Limitations

- The linear regression analysis does not establish causality. It shows that the variables are linked, but it cannot say with certainty that poverty specifically causes low literacy rates. There could be, and likely are, additional factors that influence literacy rates.
- Linear regression is a fairly simple model that cannot account for the most complex relationships. All variable relationships are assumed to be strictly linear and independent of each other. However, this simplicity also makes the model very interpretable, which made it well suited for this analysis.

Recommendations

- A comprehensive literacy improvement plan should be made at the national level
- It should include, among many other literacy improvement strategies, targeted interventions for low-income areas and interventions that attempt to address poverty directly.
- Since the report cannot prove poverty directly causes low literacy rates, the improvement plan should include frequent monitoring of literacy rates year-over-year to determine the effectiveness of these interventions and determine a plan for modifying the interventions accordingly.
- The model identified many other associated variables as well. For example, it is clear that the literacy crisis disproportionately affects certain demographic groups. As such, the literacy improvement plan should include other interventions as well, such as aiding these affected groups.

Expected Benefits

- In 1998, the average senior reading score on the NAEP was 291; by 2024, the score had lowered all the way towards 283 (National Center for Education Statistics, n.d.-a). By implementing the recommended literacy improvement plan, the trend in the score may start to reverse.
- Assuming the relationship identified by the model is in fact causal, then lowering the poverty rate by 2% would result in an increase of 1 point in the literacy score on average. (This can be gleaned by converting the 2% change onto a standardized scale, and multiplying by the model's coefficient.)

References

- National Center for Education Statistics. (n.d.-a). *NAEP Data Explorer: Grade 12 reading assessments [Data set]*. U.S. Department of Education. Retrieved April 25, 2026, from <https://www.nationsreportcard.gov/ndecore/xplore/NDE>
- National Center for Education Statistics. (n.d.-b). *The Nation's Report Card (NAEP)*. U.S. Department of Education. <https://nces.ed.gov/nationsreportcard/>
- National Center for Education Statistics. (n.d.-c). *Program for the International Assessment of Adult Competencies (PIAAC)*. U.S. Department of Education. <https://nces.ed.gov/surveys/piaac/>
- National Assessment Governing Board. (2024). *NAEP reading results: 2024 Grade 12*. U.S. Department of Education. <https://www.nagb.gov/naep/reading.html>
- Programme for the International Assessment of Adult Competencies (PIAAC). 2023. *Skills Map [Data set]*. U.S. Department of Education. Retrieved April 3, 2026, from <https://nces.ed.gov/surveys/piaac/skillsmap/>
- Yiu, T. (2020). *Understanding Linear Regression*. Towards Data Science. <https://towardsdatascience.com/understanding-linear-regression-94a6ab9595de/>